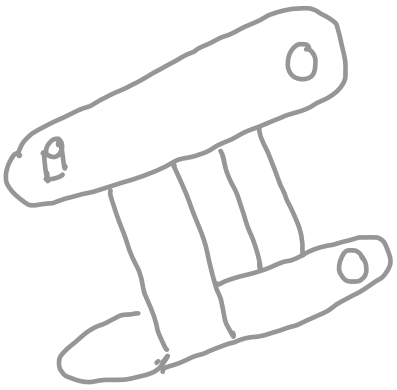
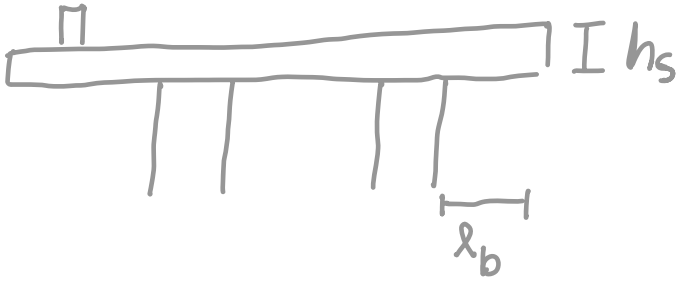
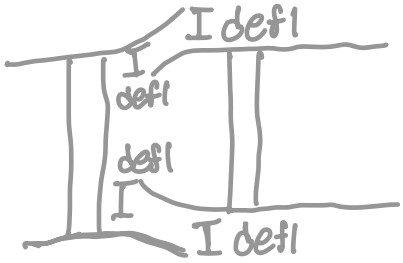




deflection across both sides, upright equidistant from both sides so equal bending from all members



knowns

resin: gray V5

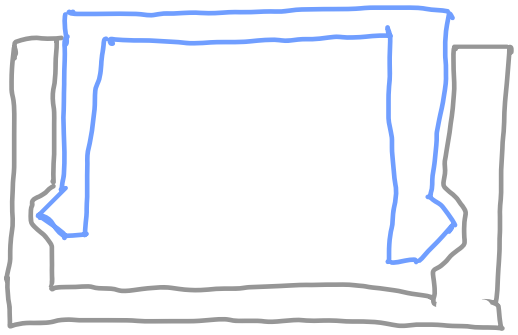
σ_u	62 MPa	ε_{allow}	65 % yield $\approx$ 1.5 %
E_t	2675 MPa	ν	.3
E	2750 MPa	$\alpha_{lead\ in}$	30°
		defl. constant	.67
ε_y	$\frac{\sigma_u}{E_t} = 2.32\%$		

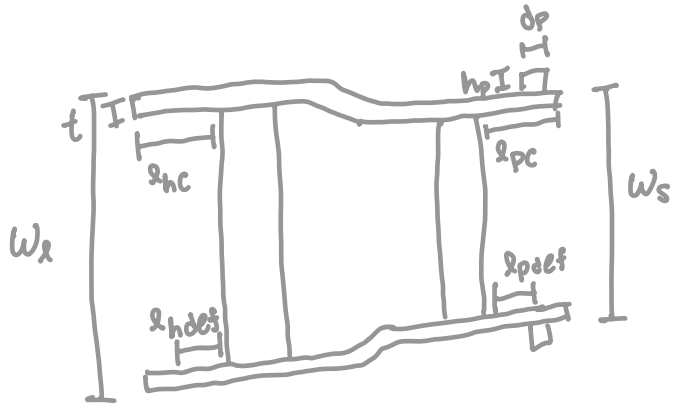
scaling igus 10-3-038-0 by 60%

w_{inside}	22 mm
h_{inside}	11 mm
pin to pin dist.	15 mm
fixed cross bar thickness	2 mm
plate thickness	1.4 mm
plate height	13 mm
inner outer plate tol.	.2 mm
d_{pin}	3.6 mm
d_{hole}	4 mm
h_{pin}	.7 mm
lead in height	.6 mm
radius of root fillet	.6 mm

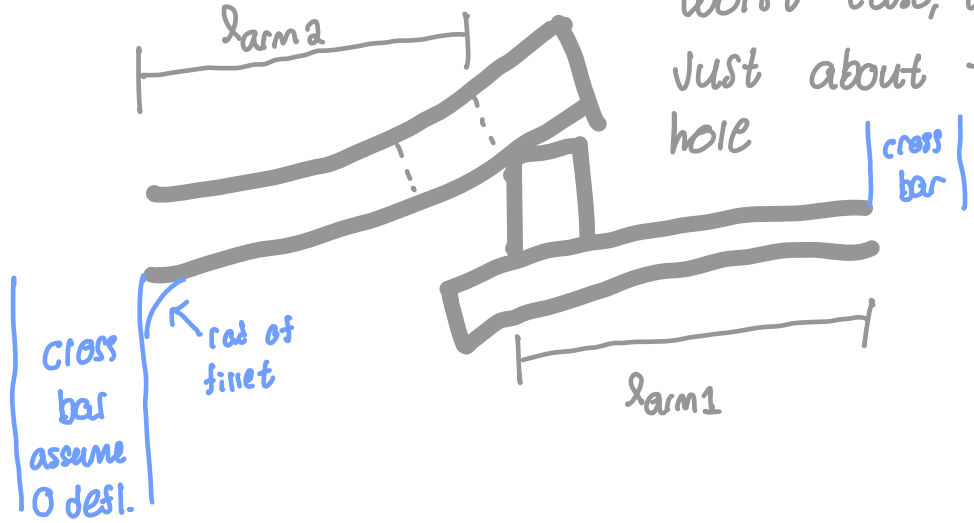
second snap fit, cover to allow for easy cable insertion

l_{arm}	7 mm
$t_{at\ mate}$	1 mm
$t_{at\ tip}$	.5 mm
undercut	.4 mm
w	4 mm





$$defl = \frac{(W_z - 2t) - (W_s + 2h_p) - 2(tol)}{2}$$



worst case, when pin is just about to meet the hole

$$l_{arm} = \frac{P2P\ dist}{2} - \frac{t_{fixed\ CB}}{2} = 7.5 - 1 = 6.5\ mm$$

$$l_{arm2} = \frac{P2P\ dist}{2} + \frac{d_{hole}}{2} - \frac{t_{fixed\ CB}}{2} = 7.5 + 2 - 1 = 8.5\ mm$$

$$\delta_1 + \delta_2 = \delta_{total} = h_{pin} - tol. = .7 - .2 = .5\ mm$$

$$\frac{\delta_1}{\delta_2} = \frac{l_{arm1}^3}{l_{arm2}^3} = \frac{274.6}{614.1} = .447 \Rightarrow \delta_1 = .5 - \delta_2,\ \delta_1 = .447\,\delta_2$$

$$.447\,\delta_2 = .5 - \delta_2 \qquad 1.447\,\delta_2 = .5$$

$$\delta_1 = .5 - .346 = .154\ mm \qquad \delta_2 = .5/1.447 = .346\ mm$$

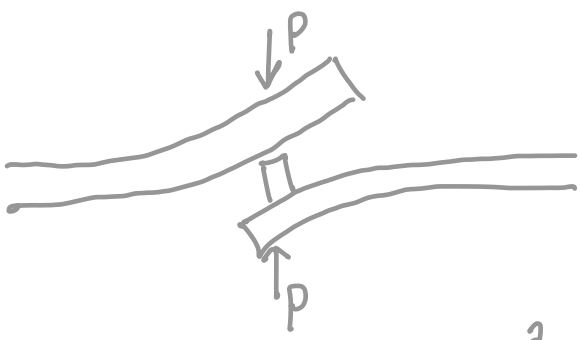
$$\varepsilon_2 = \frac{\delta_2 \cdot t}{.65 \cdot \ell_{arm2}^2} = \frac{.346 \cdot 1.4}{.65 \cdot 8.5^2} = .01028 = 1.028\%$$

$$\delta_{2\,allow} = \frac{.65 \cdot .015 \cdot 8.5^2}{1.4} = .503 \text{ mm} \qquad \frac{.346}{.503} = .688 = 68.8\% \text{ of allow}$$

$$\varepsilon_1 = \frac{\delta_1 \cdot t}{.65\,\ell_{arm2}^2} = \frac{.154 \cdot 1.4}{.65(6.5^2)} = .00785 = .785\%$$

$$\delta_{1\,allow} = \frac{.65 \cdot .015 \cdot 6.5^2}{1.4} = .294 \text{ mm} \qquad \frac{.154}{.294} = .524 = 52.4\% \text{ of allowable}$$

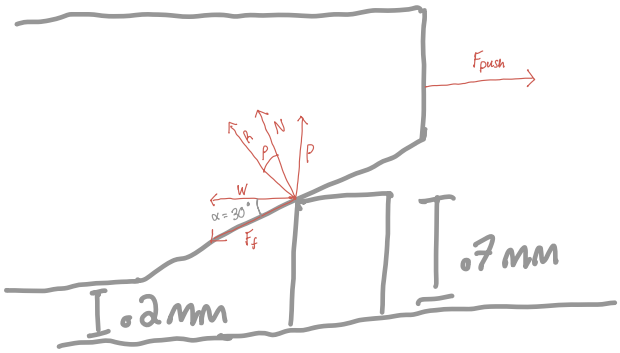
won't break, testing for insertion force now.



both plates should give equal force if we did the last part correctly

$$\text{section modulous} = \frac{\text{plate height} \cdot t^2}{6} = \frac{13 \cdot 1.4^2}{6} = 4.247 \text{ mm}^3$$

$$P = 4.247 \cdot \frac{E\,\varepsilon_2}{\ell_2} = 4.247 \cdot \frac{2750 \cdot .01028}{8.5} = 14.124 \text{ N}$$



$F = W = P\tan(\alpha + \rho)$

$P = N(\cos(\alpha) - \mu\sin(\alpha))$

$W = N(\sin(\alpha) + \mu\cos(\alpha))$

$$fm = \frac{W}{P} = \frac{\sin(\alpha)+\mu\cos(\alpha)}{\cos(\alpha)+\mu\sin(\alpha)} = \frac{\mu+\tan(\alpha)}{1-\mu\tan(\alpha)}$$

$\mu = \tan(\rho) \Rightarrow \rho = \tan^{-1}(\mu) = 16.7°$ $\quad fm(\alpha) = \tan(\alpha+\rho)$

$fm(30°) = \tan(46.7) = 1.061$

$W = P \cdot fm(30°) = 14.124 \cdot 1.061 = 14.985\ N$

$F_{push} = 2W = 29.97\ N = 6.74\ lbf$ sounds okay

technically not seperable through purely pulling force, will have to be pried apart first.